Bio-Sync: A Smart Auto-Control System for Automated Substrate Moisturizing in Mushroom Cultivation

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A Capstone Project Presented by:

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CHAPTER 1

INTRODUCTION

Background of the Project

Mushroom growing has become a significant agricultural business in the Philippines, specifically in Davao del Norte. The demand for mushroom, specifically Oyster mushroom continue to grow. Despite the significant growth, local mushroom farmers still rely in manual cultivation, especially in Panabo City. This creates a gap and high exposure to microclimate harm that can directly affect crop yield and quality. Stated in the Philippine Statistics Authority that the production of mushroom reached up to 11,800 metric tons in 2022, wherein Mindanao became one of the substantial contributor [14].

Bio-Sync is an IoT-based environmental monitoring and closed-looped control system built to maintain the temperature and relative humidity inside the mushroom fruiting chambers within the safe threshold of mycelium. The system uses DHT22 sensors that fetch real-time data with automated actuator responses to ensure consistent microclimate stability for local mushroom growers. Using smart sensors and automated IoT systems is a proven method for catching environmental stress early, making cultivation efficient and reliable [1]. This technology automatically adjusts the climate and maintains substrate moisture levels in real time, significantly differ from a manual monitoring setup that reacts to problems after they occur [2]. By making food production smarter and efficient in reducing waste, Bio-Sync directly supports Sustainable Development Goal (SDG) 2 (Zero Hunger), particularly Target 2.3 on improving the productivity and income of small-scale food producers, and SDG 12 (Responsible Consumption and Production), particularly Target 12.3 on reducing food loss along production chains. By automatically correcting temperature and humidity deviations before they cause crop failure, and by activating pumps and fans only when needed rather than on a fixed schedule, Bio-Sync helps small-scale growers protect their yields while reducing unnecessary water and electricity use. Real-time sensors help detect potential threats before they ruin a crop, cutting down on losses. Upgrading traditional

fruiting chambers with this kind of automated technology helps local growers improve their operations and boost their income [3].

The significance of environmentally-managed automation can be observed by analyzing Pleurotus ostreatus' sensitivity. According to research findings, oyster mushrooms need proper environmental conditions with temperature ranging between 24°C and 28°C and humidity between 80% to 95% [9][10]. A producer that lets either one of these variable slip will ultimately suffer the consequences of mycelial stress, either by not being able to produce mushroom pins or mushrooms that come to maturity aren't fully developed. Davao del Norte is characterized by variations in temperatures based on the season and dry seasons, thus showing that it's not simply enough to stay in the proper range—one has to work to stay there.

A dependable system is necessary especially for locations that experience sudden weather changes and drought conditions. In this case, mushroom farmers in the Philippines, whether big or small, should implement modern technology that can adapt to sudden changes in temperature and methods of drying the substrate [5]. Due to the erratic nature of tropical climate patterns, automated monitoring can be employed to avoid stress being experienced by the mycelium and thus allowing its growth [4]. In general, agricultural engineering studies support this idea, as cloud technologies demonstrate good performance even when the weather becomes unpredictable [6]. In fact, the introduction of smart devices for microclimate control in the context of smallholders contributes to better resource allocation and resilience to climatic conditions over time [7].

Bio-Sync is based on both the requirements at the institutional level and national-level research priorities such that it is not merely a response to field requirements. In regard to institutional aspects, it follows the DNSC RDE Agenda 2026-2030 (DNSC SAILS) [15] in a way which connects it to the innovative approach to modernizing main sectors of the region according to the Accelerate Agri-Aqua pillar and the ICT4D approach provided by the Innovation and Inclusion pillar that stresses the importance of using

digital instruments for improving grassroots community development. In regard to national level, Bio-Sync corresponds to CHED's A.C.H.I.E.V.E. Agenda 2025-2030 [16] under the Harmonized SDG-Based Research and Innovation pillar, at the same time sticking to the points put in DOST's Harmonized National Research and Development Agenda (HNRDA) 2022-2028 [17] emphasizing the importance of smart farming and IoT in crop production within Agriculture 4.0. All these factors emphasize that Bio-Sync is not a capstone project alone but has wide applicability and importance in both the academic and national fields.

Many local facilities are still missing round-the-clock monitoring despite facing such serious issues, but this creates a blind spot which results in high mushroom mortality and wasted resources. Although the small community grow houses in Davao del Norte are great for food production, they mostly depend on manual monitoring, working on solving humidity, and substrate problems only after trouble strikes. To achieve the best production and the health of mushrooms during growth, it is necessary to have the growth situation under constant observation. Using smart monitoring devices is proven to be a reliable solution for detecting troubles with the environment, which makes the cultivation process more convenient and effective [1].

With an increasing demand for high-value mushrooms grown in Panabo City, local facilities need a better solution for organization of the growing process than what they currently have. Nowadays, the majority of growers start acting in response to troubles. They only react to the moment when mushrooms suffer from heat or humidity stress, which often means that the damage is done already. With too much reliance on manual labor and just visual observation of growing bags, growers simply can't keep up with all the changes occurring in the environment.

Performing temperature and moisture measurements manually makes it hard to catch stress symptoms in a timely manner, especially in larger growing areas. Even if some farms use basic smart monitors, these devices may only alert farmers to the issues without solving the problems automatically. Small fluctuations in moisture levels or small

temperature increases are incredibly easy to miss. Unless growers are equipped with a system that is capable of monitoring the environment and making adjustments immediately, they will lose opportunities to prevent rest harvest failures.

The proposed way to resolve this problem is through the development of Bio-Sync, an automated system that is able to monitor and control microenvironments in mushroom fruiting chambers. This system will use a network of sensors –the DHT22 which will measure temperature and humidity in real-time. In case the temperature rises above or humidity goes below the established limits, Bio-Sync will automatically correct the environment without waiting for human actions. The system's efficacy will be assessed according to the accuracy of sensors, automatic response, and the overall effectiveness of ensuring stable growing conditions.

Objectives of the Study

The main objective of this study is to build an autonomous mycological monitoring and smart auto-control environmental system, Bio-Sync, that uses integrated IoT architecture to maintain optimal microclimates in local mushroom fruiting chambers. Specifically, the researchers aim to:

1. Collect the ambient temperature and relative humidity inside mushroom fruiting rooms with the help of an IoT sensor network built using DHT22 sensors combined with a NodeMCU ESP8266 microcontroller for continuous real-time data monitoring.
2. Compare and implement three machine learning algorithms — Decision Tree, Random Forest, and K-Nearest Neighbor (KNN) through: dataset preparation, feature engineering, model training, validation, and performance evaluation based on accuracy, precision, and recall, to classify the fruiting chamber's environmental conditions (normal, warning critical) and deploy the most effective model within the system.
3. Implement an automated hardware response system using classified environmental conditions and predefined ideal mycelial growth parameters

to activate misting pumps or ventilation fans and automatically correct deviating conditions without human intervention.

4. Create a centralized online dashboard for real-time monitoring and analysis of past trends and remote management of data.

Significance of the Study

- The intention for this research is to transfer mushroom cultivation from a manual operation where mushroom producers react to issues into a method of growing mushrooms where the grow room does most of the work automatically. The use of modern sensors will be the basis for mushroom cultivation that will be more efficient and cost-effective, thus moving local agriculture forward. This project will primarily be helpful to the following parties:
- Mushroom Farms in Panabo City: As a result of the implementation of this technology, local producers will be able to minimize the number of ruined or infected crops, manage growing processes efficiently, provide huge savings on energy and water consumption, and deal much easier with climate change.
- Mushroom Farmers: It means that mushroom growers will be able to save time since the system automatically controls the climate and they won't have to check the bags every once in a while. The system will work automatically and eliminate problems if the temperature is too high or the substrate is too dry. Moreover, this technology allows mushroom producers to rely on scientific data for making decisions instead of guessing.
- Local Workers: Workers will have a less exhausting job since they will be relieved from the burden of manual misting of mushrooms and checking thermometers as the new system will take emergency steps to regulate temperature automatically.

- Future Researchers and Developers: The endeavor gives a practical reference point for future research about automated mushroom cultivation in community environments. The information and data obtained here will be useful for the introduction of new technologies and agricultural practices, which can improve not only Panabo City but also other farming municipalities wishing to upgrade their facilities.

Scope and Limitations

The aim of the research is to create and evaluate the effectiveness of the Bio-Sync device for the control of climate and moisture in the local fruiting chambers of oyster mushrooms. The DHT22 sensors are used for tracking continuously the temperature of the environment (−40°C to +80°C and the precise measure of the temperature, which is ±0.5°C) as well as the RH level (0% – 100% RH and the accuracy of ±2% to 5%). The information from the sensors is evaluated by the NodeMCU ESP8266 microcontroller, programmed with custom limits for each stage of the development of mushrooms that are known as fruiting (24°C to 28°C and RH – 80% to 95%), colonization (24°C to 28°C and RH – 70% to 85%), and for the pinning initiation (18°C to 24°C and RH 85% – 95%).

However, there are some real limitations to this study. The tests used here focus on one fruiting compartment only, meaning the system has not been tested in various compartments at once. Also, the machine-learning module is trained only based on the data collected in the Bio-Sync and does not include the outside mushroom cultivation datasets, which might limit the general applicability of the system. The technology being researched is reliant on a continuous local Wi-Fi connection: otherwise, the system would switch to fail-safe mode, stopping the logging process briefly. Moreover, the method used here measures humidity through DHT22 measuring the humidity indirectly. Temperature and humidity are among the major studied variables of the environment when it comes to growing Pleurotus ostreatus.

Review of Related Literature and Works

The amalgamation of IoT systems in mushroom cultivation is an important development in current agritech mainly because it solves a particular problem – crop loss caused by variable microclimates [1][2]. Regular mushroom farming relies too much on manual observation and subjective assessment of temperature and humidity which results in subpar yields and slow response to environmental pressure. Othman et al. [1] discovered that the use of IoT-based environmental monitoring devices for mushroom growing at home improve stability of the environment and significantly reduce loss of crops associated with human error. Similarly, Kailashkumar et al. [2] arrived to the same conclusions, however, their focus was on developing the system for the cultivation of oyster mushrooms, which allowed them to demonstrate that automated monitoring leads to more controllable output. In both national and international contexts, mushroom management has moved considerably away from simple monitoring towards proper actuator platforms [3].

According to Singh et al., have made a similar point about the shift from reacting to climate challenges to adopting an effective method of managing climate [4]. Their idea emphasizes that there is more to IoT-enabled smart agriculture than a mere network of sensors — it is necessary to have advanced environmental control systems that can actively react to problems even before a fungus is under stress. In this context, modern solutions — such as smart growing kits [5] and systems based on fuzzy logic algorithms [8] — may further define closed-loop systems. What is particularly important about fuzzy techniques is that it makes possible distinguishing different degrees of temperature and humidity imbalance, rather than categorizing them as either meeting the required values or not. This, in turn, allows for more sensitive action in terms of activating ventilation fans and water pumps, for instance, which is impossible in case of a binary system. What is also stated in more recent research is the primary role of precise temperature and humidity regulation in mushroom production [9][10].

Because environmental thresholds differ throughout a mushroom's life cycle, stage-specific limits regarding optimal conditions need to be established as a priority in

a system for the automated cultivation of mushrooms. Oyster mushrooms are particularly sensitive to microclimate, and “optimal” conditions during the three main growth stages are not the same.

At the stage of colonization, mycelium grows through the substrate it inhabits, and it must be kept within the range of 24°C–28°C with the relative humidity of 70%–85%. If the temperature rises above 30°C, the growth will be affected, but the biggest threat is thermophilic contamination that may occur.

For the pinning stage, though, the required parameters change. The temperature must be kept at 18°C–24°C, while the relative humidity must be increased to 85%–95%.

Finally, during the fruiting phase, the temperature will change again and should be kept in the range of 24°C–28°C and the relative humidity of 80%–95%.

Table 1. Environmental Thresholds for Pleurotus ostreatus Cultivation Stages

Growth Stage	Temperature Range (°C)	Relative Humidity (%)	Key Reference
Colonization	24°C – 28°C	70% – 85%	[2], [9]
Pinning Initiation	18°C – 24°C	85% – 95%	[10], [1]
Fruiting (Active)	24°C – 28°C	80% – 95%	[1], [9], [10]

These established thresholds aren't just merely numbers — these are the very numbers that Bio-Sync uses in its microcontroller. These numbers are not arbitrary; Bio-Sync’s automated responses are not mere arbitrary values, but instead stem from research-based data that is stage centered.

To comprehend the reasons behind the movement of these thresholds, it is crucial to investigate their biology.

Stage 1, colonization, starts immediately after the substrate was implanted with spawn. Here, mycelium begins its expansion inside the organic material, and growing bags are kept constant so that CO2 level is held high and it is protected from contamination. Temperature rises above 30 degrees Celsius and humidity level disturbs below 70% RH [2][9].

Stage 2, which is the beginning of the pinning phase, can be initiated only after the colonization stage has been completed. During this stage the environmental conditions are manipulated in order to force the mushrooms into fruiting: the substrate bags are opened, fresh air enters the mushrooms' growing area, the temperature is set at 18°C to 24°C, and humidity is increased to 85% to 95% RH. This is the most sensitive stage of the process done during the whole life cycle of mushrooms, and any mistake during this stage may also mean the termination of the pinning process [4][10].

Stage 3 is the fruiting stage, which includes the formation of the caps. The temperature is kept at 24°C to 28°C and humidity at 80% to 95% RH during this stage. The airflow is also of great importance, as excess of CO2 due to not proper ventilation can result in longish stems and not developed caps instead of complete fruits. The caps become flat when the mushrooms are ripe enough for harvesting [1][6][9].

Bio-Sync employs three distinct approaches to achieve classification, namely Decision Tree, Random Forest, and K-Nearest Neighbor, while classifying the environmental conditions of the fruiting chamber from sensor data. The reason for selecting Decision Trees is that they are quite simple and intuitive for people who are not well versed in technology [13]. According to researchers Singh et al. [4], Decision Trees are useful in environmental detection for mushroom growing, with accuracy exceeding 90% in detection processes. Random Forest is another algorithm which also relies on Decision Trees but uses a group of them, thus preventing overfitting. KNN has a different approach. It is a non-parametric algorithm that involves classifying the data based on the nearest neighbors, and this method has proven to be quite effective for handling low-dimensional feature set data. Pantilanan et al. [9] provided similar results in mushroom environment classification through the use of KNN.

The three models all use sensor log data that is obtained directly by Bio-Sync using a five-step procedure, which involves collecting raw data from the DHT22 sensors while performing other steps like extracting relevant features, labeling data according to certain thresholds, splitting the data for training and testing, and conducting cross-validation. Performance is evaluated using different metrics, such as precision, recall, accuracy, and F1-score, in order to determine the best-performing model for deployment on the system.

The DHT22 (AM2302) is a highly accurate temperature sensor that is widely used in IoT installations for agricultural monitoring. It measures temperature across the ranges of −40°C to +80°C with accuracy of ±0.5°C and humidity across the range of 0%–100% with accuracy of ±2% to ±5%, with a sampling rate of two seconds. Since Othman et al. [1] and Kailashkumar et al. [2] have both used DHT22 sensors in their projects on mushroom cultivation, they have confirmed that the sensor provides the required level of accuracy for microclimate management.

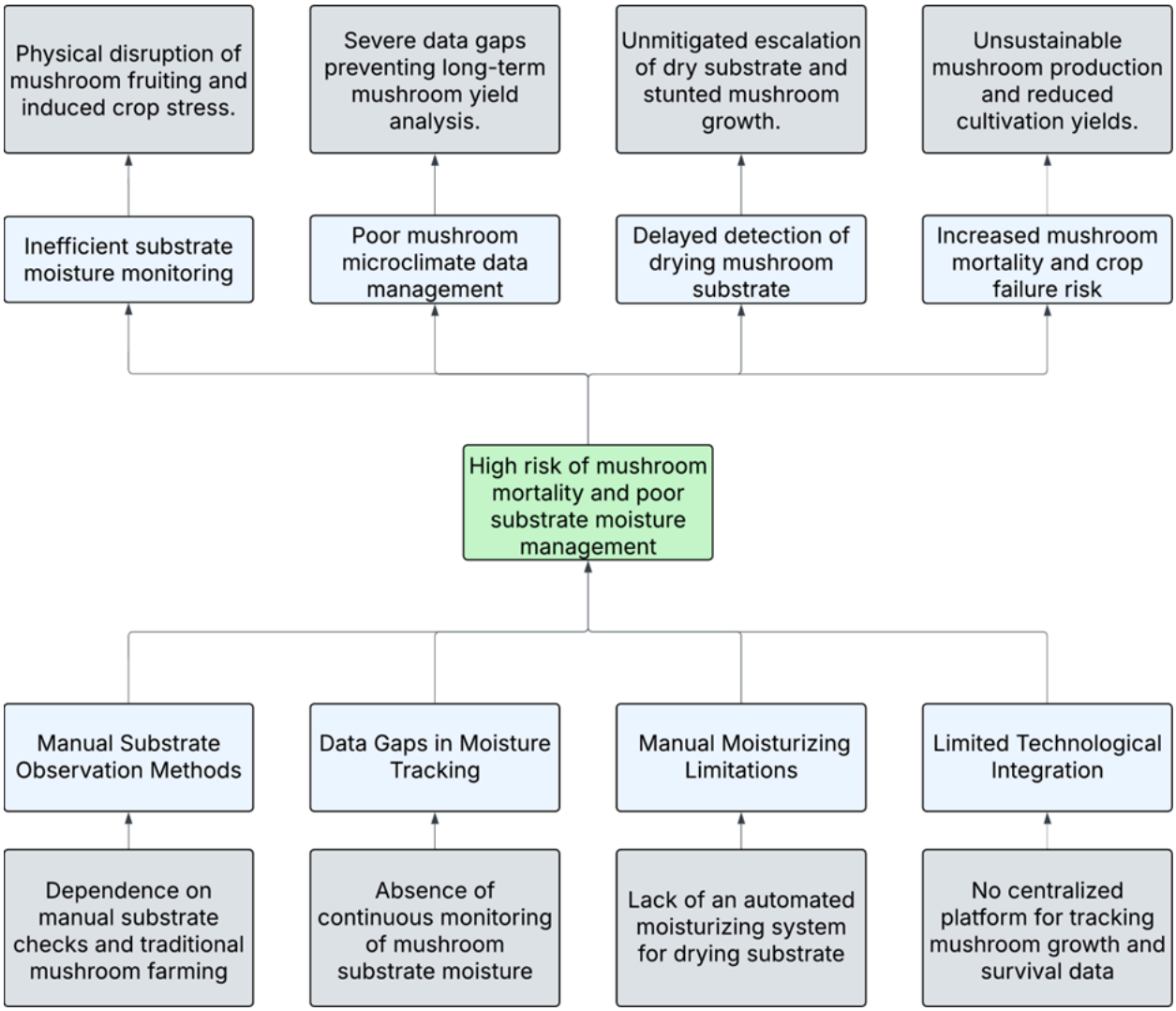


Figure 1: Problem Tree Analysis

Discussion: As shown in the Problem Tree Analysis, the cause and effects of the problem of mushroom cultivation in Panabo City can be analyzed using this diagram. The tree trunk stands for the main problem which is high death rate of mushrooms and erratic yields. The tree roots symbolize the major causes of the problem such as the absence of round the clock monitoring, delayed response to drying of substrate and reliance on manual monitoring of climate. The branches signify the negative effects caused by the problem like erratic supplies in the market, financial losses for local farmers and waste of resources.

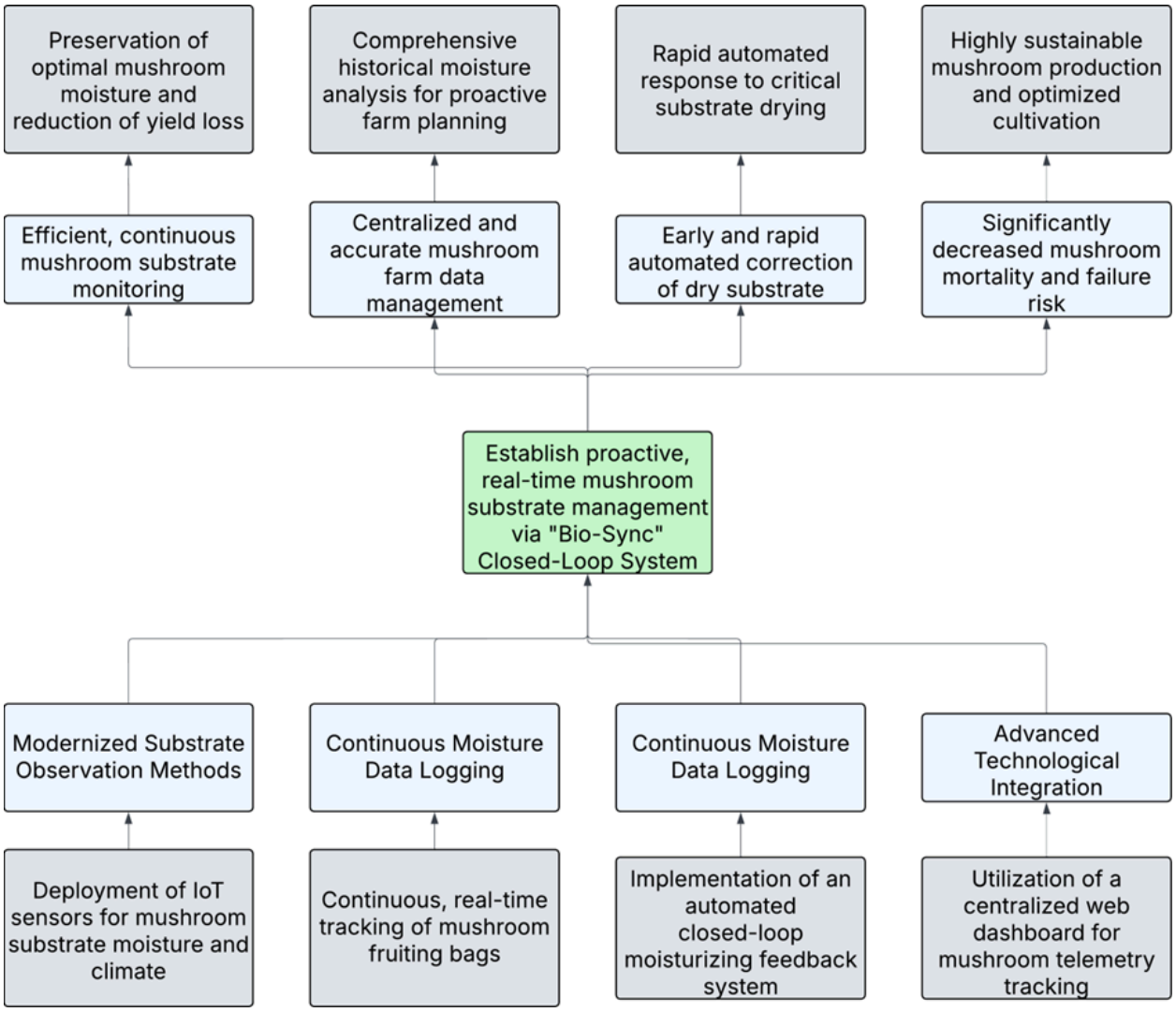


Figure 2: Objective Tree Analysis

Discussion: The Objective Tree represents the solution for the problem illustrated in the previous tree diagram. The tree roots symbolize the actions needed such as installing an IoT sensor network and setting up the closed-loop feedback system. The tree trunk represents the goal to be achieved and that is to maintain an optimal microclimate automatically. The branches show the positive impact of the solution like the maximized yields, minimized labor, optimized use of water and stable supply of mushrooms in the local market.

Table 2. Comparative Analysis of Bio-Sync and Related Systems

Features	System 1: IoT-Based Environmental Monitoring System for Home-Based Mushroom Cultivation [1]	System 2: IoT-Enabled Smart Cultivation System for Oyster Mushroom Production [2]	System 3: Automated Mushroom Monitoring and Management System [3]	Proposed System (Bio-Sync)
24/7 Environmental Monitoring (Temp, Humidity, Moisture)	✓	✓	✓	✓
Centralized Web Dashboard		✓	✓	✓
Basic Automated Actuation (Pumps & Fans)		✓	✓	✓
True Closed-Loop Feedback (Dynamic Self-Correct				✓

ion)				
Alert and Notification System				✓
Web-Based Data Visualization	✓		✓	✓
Remote Access				✓

Discussion: Table 2 shows that even though the existing methods satisfy basic requirements of IoT monitoring and actuation, the idea of Bio-Sync is better since it offers complete closed-loop feedback with dynamic self-correction and complete integration of the web-based visualization tools

Definition of Terms

Actuation: refers to the mechanism which allows the system to respond in a physical way to the digital instruction, i.e. for example, switching the misting pump or the ventilation fan to change the conditions in the surrounding environment.

Bio-Sync: is the name of the proposed capstone project which is a fully automatic closed loop control system for regulating the conditions necessary for oyster mushroom growing.

Smart Auto-Control System: is the control system which has a continuous input monitoring and output controlling (actuator action) of the system in order to keep a specific state without any human presence.

DHT22 Sensor: is a kind of temperature and humidity digital sensor (working in the range from –40 C to +80 C, with 0.5 C accuracy; measuring humidity from 0 to 100%) used in the framework of the research to check temperature and humidity in the fruiting chamber.

Internet of Things (IoT): refers to the system which consists of the physical objects equipped with the sensors and software programs which allow them to collect data and share it via Internet.

Microclimate: is the specific climate inside the mushroom fruiting chamber which differs from the climate on the surface.

Mycelium: is the part of the fungus, representing the vegetative body of the fungus with an effective structure of filaments (hyphae). This organism is very sensitive and it is necessary to monitor the environment strictly before and during the fruiting process.

Pinning / Primordia: is the first visible step in the process of the appearance of the fruiting body when there are small forms of mushroom appear if required temperature is reached and humidity is high enough.

Pleurotus ostreatus: is the Latin name for the oyster mushroom which is considered as the main species under study in the framework of the Bio-Sync project.

Substrate: The bulk organic material (often sawdust, rice straw, or agricultural waste) contained in growing bags, which serves as the food source and physical base for growing mushrooms.

CHAPTER 2

METHODOLOGY

The systematic method for planning, developing, and executing the Bio-Sync project is discussed in this chapter. The procedures followed to identify system specifications, evaluate hardware/software performance, design the closed-loop architecture, create testing and deployment protocols are explained. Each of these phases relates to the purpose of the Bio-Sync system, which is to ensure the maintenance of stable temperature and humidity in mushroom fruiting rooms using automated IoT control.

The implementation of the Agile Software Development Life Cycle alongside the PADIM framework (Planning, Analysis, Design, Implementation and Maintenance) was used as a guideline for achieving the Bio-Sync architecture. PADIM was chosen to be the overall structure since these five consecutive steps naturally fit into the solution steps of a hardware-software capstone project from the site assessment and sensor acquisition to the field implementation and maintenance. Agile was implemented as a part of this structure rather than in place of it, because the project needs constant iterations in hardware adjustments, threshold changes and software corrections based on the information obtained through live sensors. Thus, PADIM determines what should be done at each stage, while Agile determines ways of adjustments at each of these stages until achieving the required performance level.

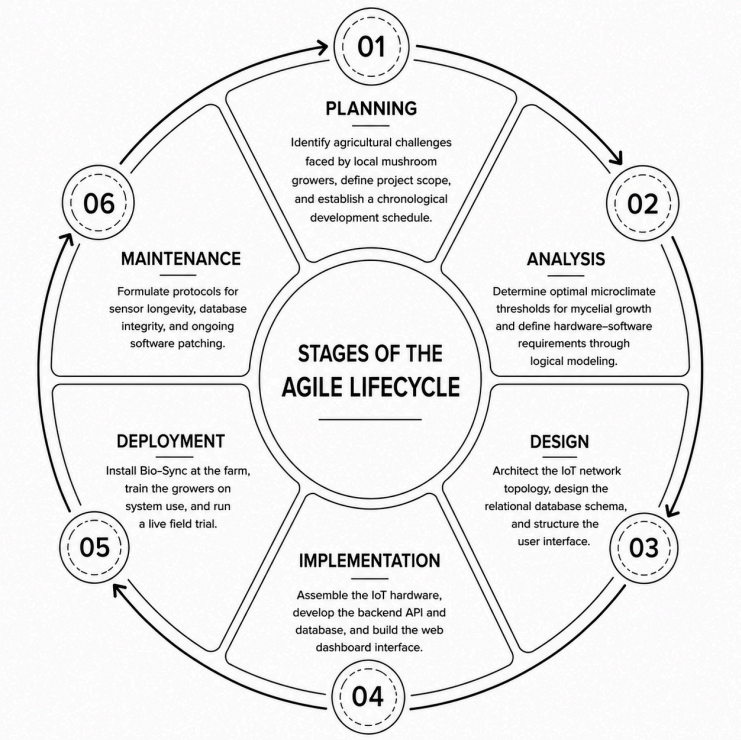


Figure 3. Graph of the PADIM-Agile Model Phases

Discussion: Figure 3 lays out how the PADIM framework is combined with the Agile methodology. It demonstrates how the hardware and software work together in processes that overlap and iterate, which makes it possible to perfect the closed-loop control system based on real-time sensor data received.

System Planning

In this part, the reader can understand how the group conceived the Bio-Sync project for tackling the problem of high mortality rate of mushrooms that arises due to the absence of round-the-clock monitoring at local facilities. Here is where the basic framework was created in terms of time management and organization of the team.

Project Team Organization

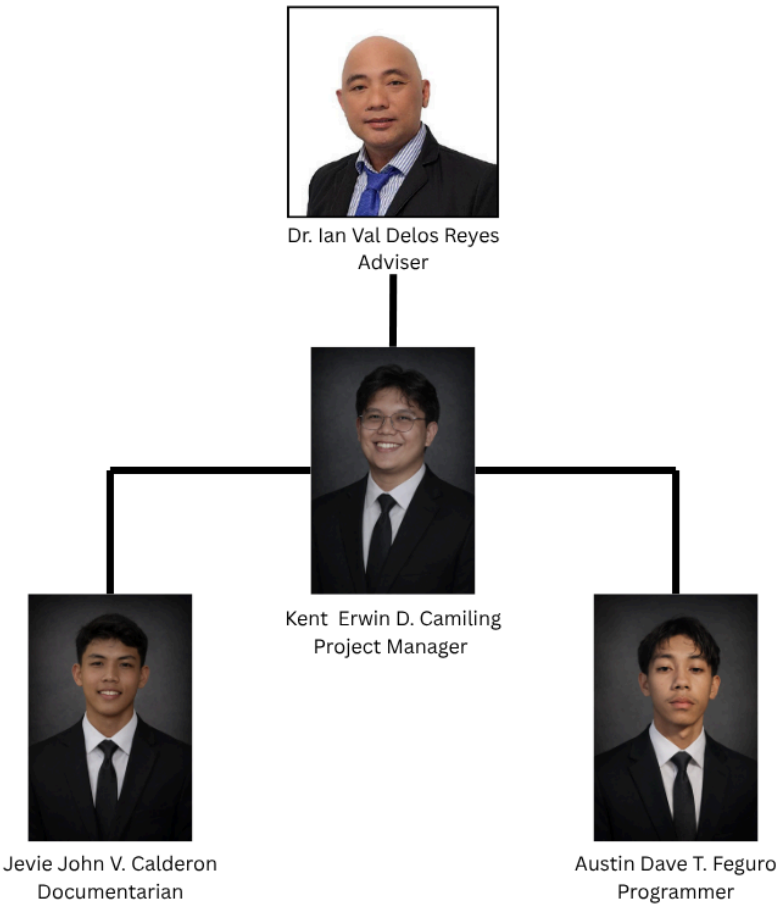


Figure 4. Project Team Organization

Discussion: This figure represents the organizational structure of the capstone team. The figure describes the responsibilities of each member of the team, namely Dr. Ian Val DeLos Reyes, Kent Erwin D. Camiling, Austin Dave T. Feguro, and Jevie John V. Calderon.

Word Breakdown Structure (WBS)

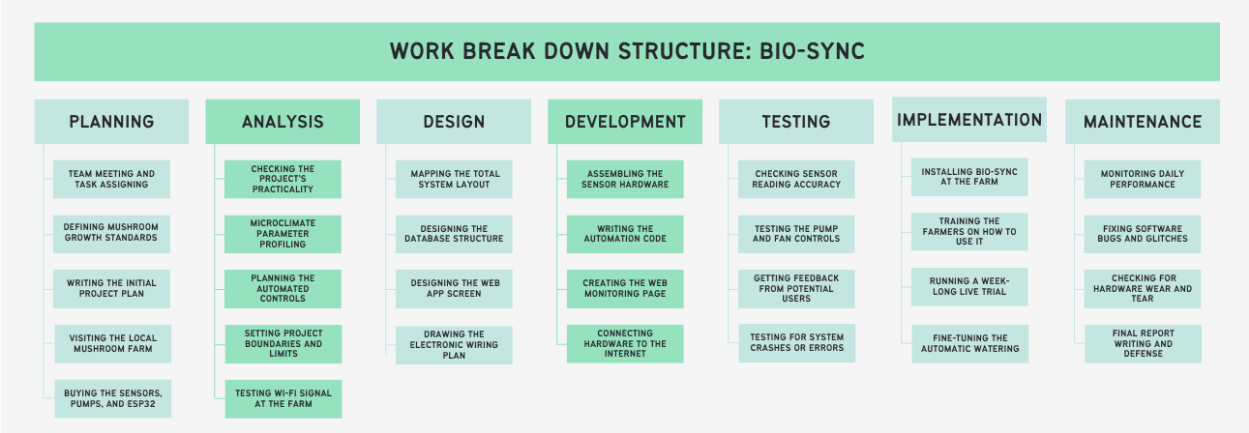


Figure 5. Work Breakdown Structure

Discussion: This structure effectively breaks down the entire Bio-Sync project into different tasks, categorized into various PADIM phases, starting from site assessment and purchase of hardware up to backend coding, integration of the system, and deployment.

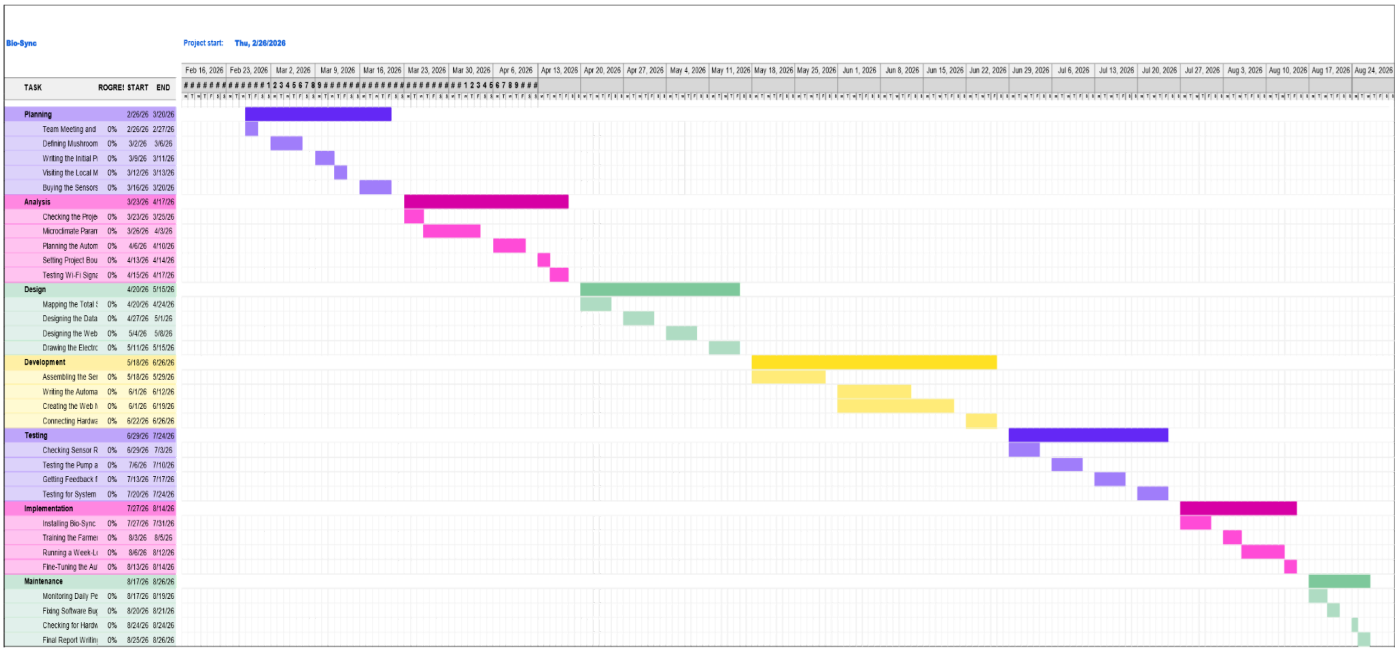


Figure 6. Gantt Chart

Discussion: Gantt Chart provides an overview of the timeline and scheduling of the Bio-Sync project starting from its initiation all the way to the completion deadline of August 2026.

System Analysis

System Architecture

The architecture of Bio-Sync forms the basis of the system design in terms of hardware, software and data interaction. Hardware involves the use of DHT22 sensors to monitor ambient temperature and humidity as well as moisture sensors installed in the substrate. All these feed data to the microcontroller. The software involves the central web-based dashboard where data is analyzed and compared against the threshold limits. When the threshold levels are exceeded, then the architecture indicates that there is a need to respond through hardware action in form of activating misting water pumps or fan systems.

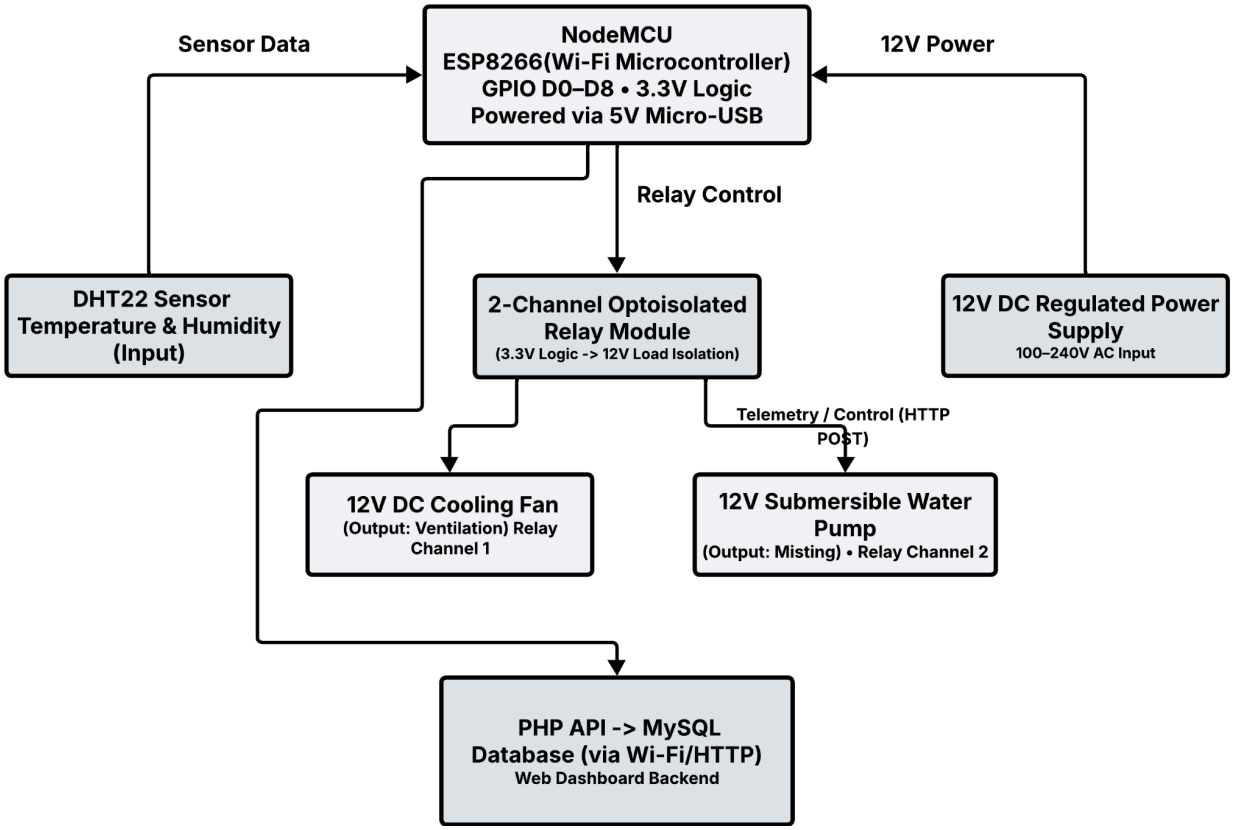


Figure 7. System Architecture

Discussion: The System Architecture shows the three-tiered architecture, which will allow us to have autonomous and closed-loop control of the process of mushrooms growing. The Hardware Tier shows two prototyping approaches: Option A includes NodeMCU ESP8266, whereas Option B consists of Arduino Uno R3 connected to the ESP-01 Wi-Fi module. DHT22 telemetry data will be preprocessed in the edge device, sent through Wi-Fi and stored in the MySQL database using the centralized PHP API. Once any environmental thresholds are exceeded, the controller will automatically trigger optoisolated relays for the 12V DC fan or water pump operation.

Functional and Non-Functional Requirements

Functional Requirements:

1. The system should receive the current values of temperature and relative humidity.
2. The system should use the best ML classification algorithm to classify the environment state.
3. The system should produce automated notifications for critical environmental changes.
4. The system should automatically turn on the water pump and fan if the readings of the environmental sensors exceed certain thresholds.
5. The system should have a web dashboard.

Non-Functional Requirements:

1. **Accuracy (Functional Suitability)** –The DHT22 sensor values will be verified with the help of a standard instrument during the calibration process.
2. **Performance (Performance Efficiency)** – The transferring of data from the sensors to the web application will happen with minimum possible latency.
3. **Compatibility** – There will be compatibility between the NodeMCU ESP8266 microcontroller and peripherals on one side and the web dashboard and database on the other. It means that there will be no problems using them together regardless of the type of the device or browser chosen by the user.
4. **Usability** – The GUI interface will be understandable and available for the agricultural workers who have low levels of technical knowledge.
5. **Reliability** – The physical system will work 24/7 without breaks.

6. **Security** –The system will limit the number of users to those who have an access through the authentication system. Also, the system will ensure the protection of the information that is stored on the server.
7. **Maintainability** – The code and hardware will be modular and easy for modification.
8. **Portability** – The system will be portable across the various fruiting chamber constructions.

Use Case Diagram

The use case diagram describes the requirements of the Bio-Sync system by specifying the key actors involved and the manner through which they interact with the automated environment and the digital interface.

The two key actors are: the Facility Owner/Operator and the Autonomous System (Bio-Sync Hardware). As the hardware performs independently to control the environment, the facility owner is the key human actor responsible for running the facility.

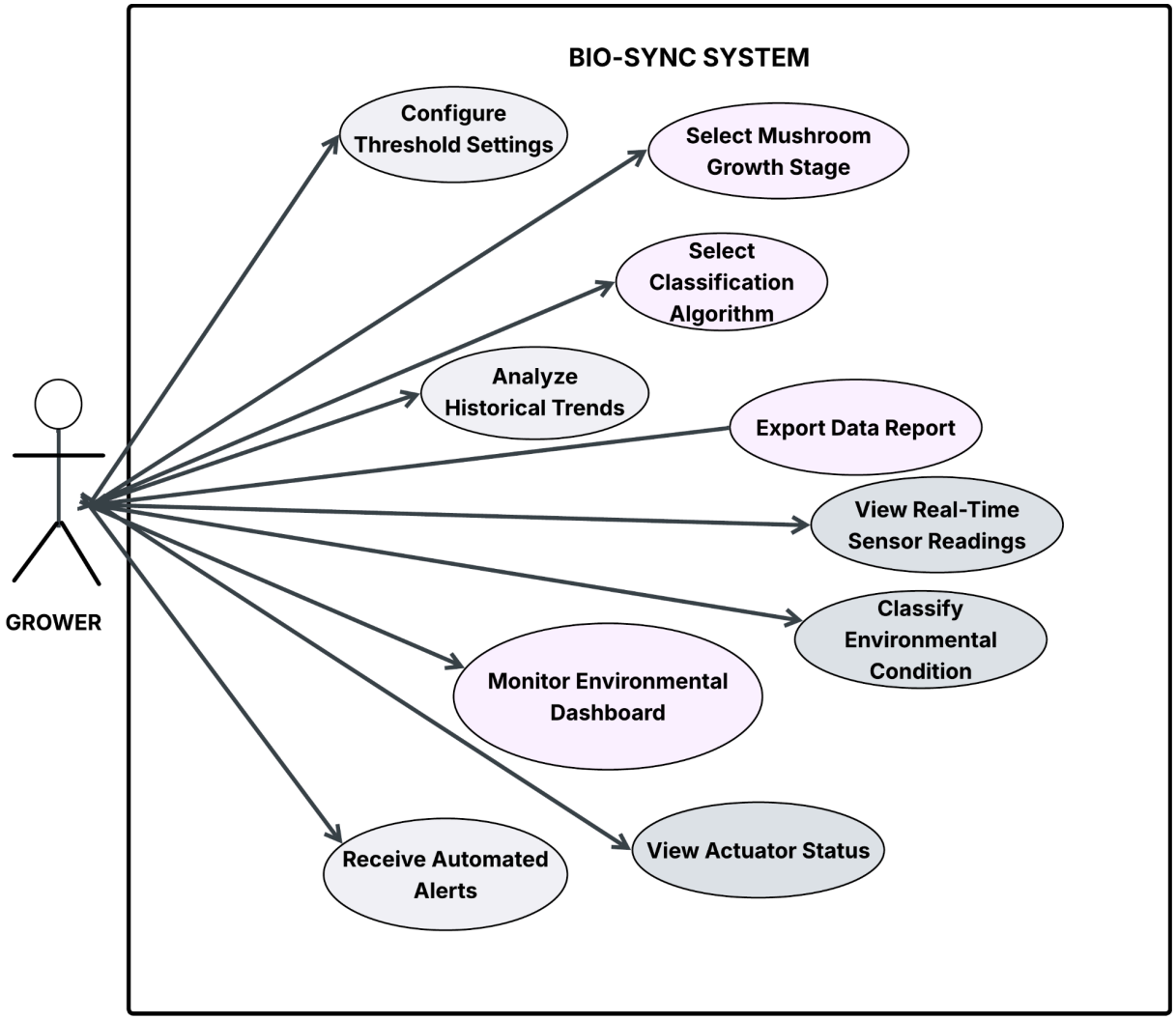


Figure 8. Bio-Sync System Use Case Diagram

Discussion: This figure presents the relationship between the Facility Operator and the Autonomous System. It depicts the process through which the hardware independently monitors the metrics and acts on the devices, and through which the user has access to the dashboard and controls threshold limits and historical data.

Context Flow Diagram

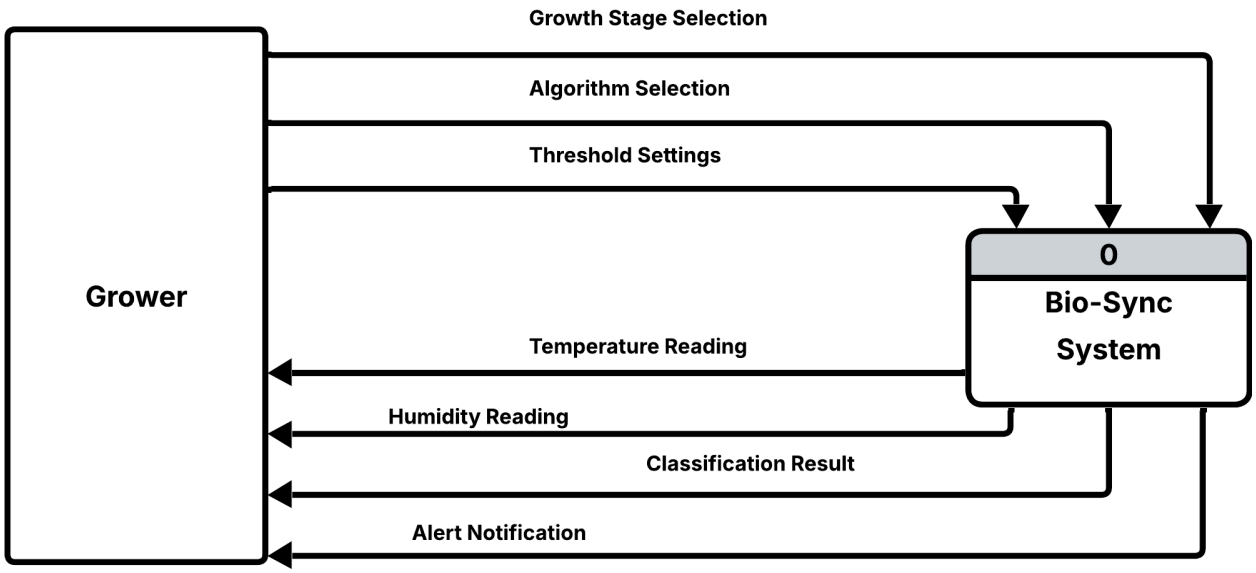


Figure 9. Context Flow Diagram (Level 0 Data Flow Diagram)

Discussion: The Level 0 DFD is the high-level description of the Bio-Sync ecosystem, indicating the data flow process whereby the environmental variables are captured by the physical sensors and then processed by the Bio-Sync processor to generate hardware and digital actions (misting/ventilation) at the user's dashboard.

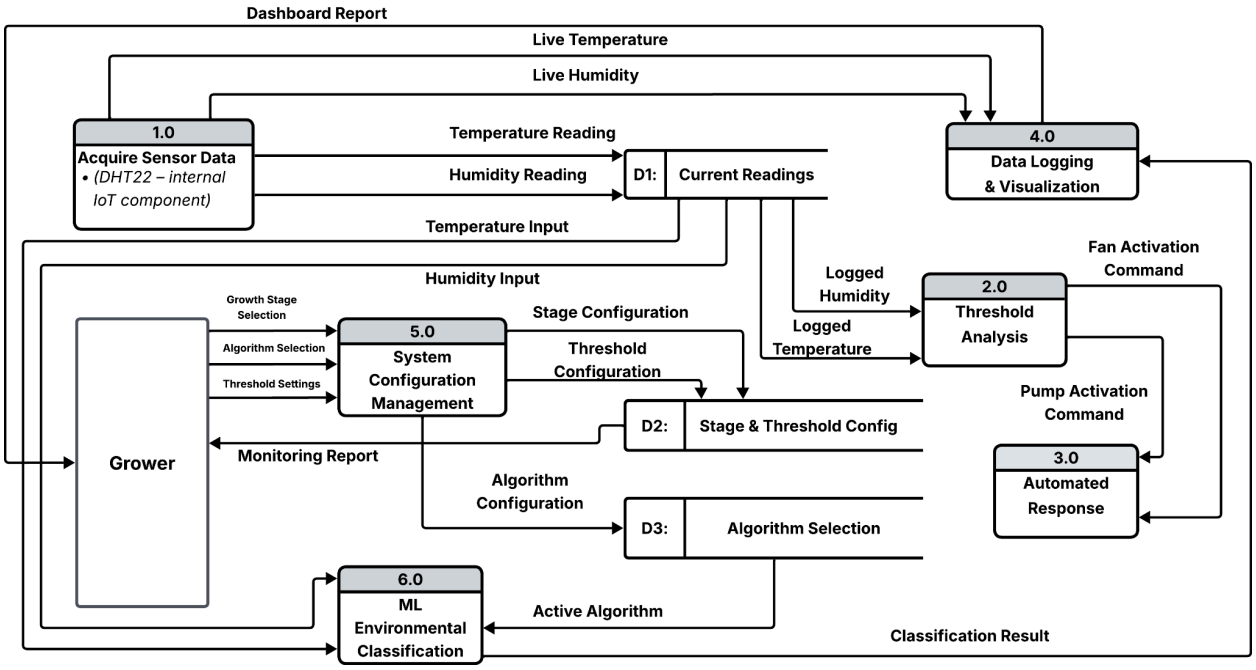


Figure 10. Data Flow Diagram Level 1

Discussion: This diagram details the high-level process into sub-processes. This provides details about the data flow; this includes data flow from the DHT22 and moisture sensors capturing raw data, data validation from the microcontroller based on the set threshold values and trigger relay for the actuators and finally push logs to the cloud database.

System Design

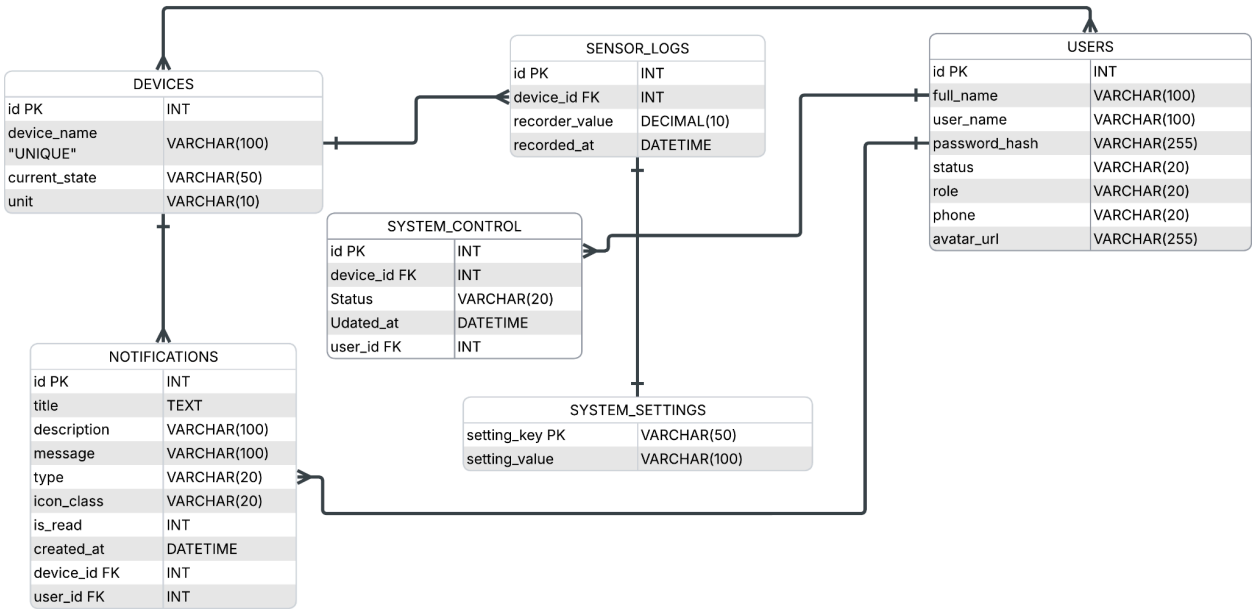


Figure 11. Entity Relationship Diagram

Discussion: The ERD is used to design the database for the project. This includes relationship between data entities such as the Users (operator) table, Sensor Logs (temperature, humidity, moisture data with time stamp) and Actuator_Logs (Pump and Fan trigger).

JSON Schema Diagram

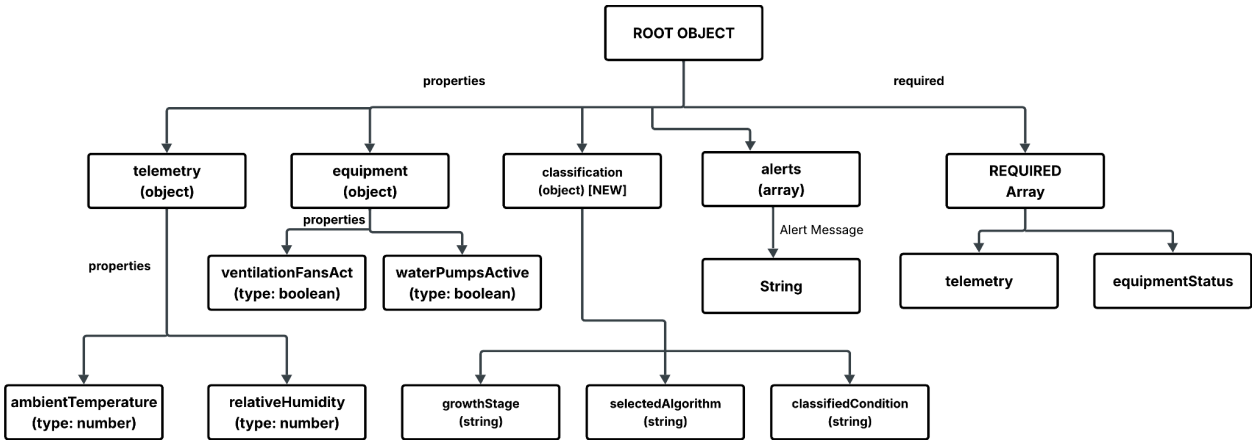


Figure 12. Schema Diagram

Discussion: The JSON structural schema is used in this diagram. The purpose is to represent how the data is transferred from the IoT microcontroller to the API of the web application and also for data standardization.

The section describes the relational structure of the database engine that will be used in the Bio-Sync web application platform.

Data Dictionary

The section describes the relational structure of the database engine that will be used in the Bio-Sync web application platform.

Table 3: Users Table

Field Name	Data Type	Constraints/Key Type	Description
id	INT	Primary Key, Auto-Increment	Unique identifier for each user account.
full_name	VARCHAR(100)	None	The complete legal name of the operator.
user_name	VARCHAR(100)	None	Unique username used for web dashboard authentication.
password_hash	VARCHAR(255)	None	Cryptographically secure hash of the user's password.
status	VARCHAR(20)	None	Account status (e.g., 'Active', 'Inactive').
role	VARCHAR(20)	None	User permission tier (e.g., 'Admin', 'Operator').
phone	VARCHAR(20)	None	Contact number for dispatching system notifications.
avatar_url	VARCHAR(255)	None	Path or URL to the user's profile image asset.

Table 4: DEVICES

Field Name	Data Type	Constraints/Key Type	Description
id	INT	Primary Key, Auto-Increment	Unique identifier for each hardware node.
device_name	VARCHAR(100)	None	Distinct descriptive name of the peripheral.
current_state	VARCHAR(50)	None	Active runtime state or mode (e.g., 'ON', 'OFF').
unit	VARCHAR(10)	None	The target measurement metrics unit handled (e.g., '°C', '%').

Table 5: Sensor_Logs

Field Name	Data Type	Constraints/Key Type	Description
id	INT	Primary Key, Auto-Increment	Unique record locator for the specific log transaction.
device_id	INT	Foreign Key (References DEVICES.id)	Links the recorded data to its originating physical sensor.
recorded_value	DECIMAL(10,2)	None	Holds the numeric sensor reading value (e.g., temperature in °C or humidity in

			%RH) captured at the time of logging.
recorded_at	DATETIME	Default: CURRENT_TIMESTAMP	Precise timestamp when the telemetry reading was captured.

Table 6: System Control

Field Name	Data Type	Constraints/Key Type	Description
id	INT	Primary Key, Auto-Increment	Unique tracking identifier for the control sequence.
device_id	INT	Foreign Key (References DEVICES.id)	Links the control command to the target hardware device.
status	VARCHAR(20)	None	State outcome of the operational action.
updated_at	DATETIME	Default: CURRENT_TIMESTAMP	Timestamp recording when the control command was last executed or modified.
user_id	INT	Foreign Key (References USERS.id)	Note: A relationship line links USERS to this table, but the attribute is missing. Recommend adding this to track which operator fired the command.

Table 7: System Settings

Field Name	Data Type	Constraints/Key Type	Descriptions
setting_key	VARCHAR(50)	Primary Key	Unique lookup configuration string flag.
setting_value	VARCHAR(100)	None	Assigned variable data limit value.

Table 8: Notifications

Field Name	Data Type	Constraints/Key Type	Descriptions
id	INT	Primary Key, Auto-Increment	Unique transaction index for the specific alert.
title	TEXT	None	Short header label summarizing the alert notification event.
description	VARCHAR(100)	None	Brief supplementary detail providing context for the alert.
message	TEXT	None	Full content body text of the dispatched alert notification.
type	VARCHAR(20)	None	Classification of the alert category (e.g., 'Alert', 'Info', 'Warning').
icon_class	VARCHAR(20)	None	CSS icon class reference used to render the

			notification indicator in the UI.
is_read	TINYINT(1)	None	Boolean state flag (0 for Unread, 1 for Read).
created_at	DATETIME	None	Timestamp showing when the system event alert was fired.
device_id	INT	Foreign Key (References DEVICES.id)	Note: A relationship line links DEVICES here, but the structural field column is missing in the box.
user_id	INT	Foreign Key (References USERS.id)	Note: A relationship line links USERS here, but the structural field column is missing in the box.

Technologies, Concepts, and Theories

Putting the whole system together meant integrating four relatively separate fields of technology into the project. These include traditional control theory to regulate how the system responds to sensor inputs, machine learning to decipher what that input means, embedded systems programming for implementation on limited hardware, and simple web-based tools to output all this information in a user-friendly way to someone who doesn't code. The four are detailed further along with the hardware they utilize.

Closed-Loop Control Theory

Fundamentally, Bio-Sync operates on a closed-loop feedback principle as opposed to a time-based system. An open-loop system turns on and off a pump every so often without taking into account the actual requirements of the growing substrate, which is the very type of dummy automation this project was attempting to avoid. Instead, the system uses the data from its own functions: temperature and humidity are measured, compared to the ideal parameters for the corresponding stage of the growth process, and only then the fan or the pump is activated. The sensing-comparing-correcting circle is repeated continuously, which means that the misting pump does not operate for a specified time period. It will run until the humidity readings show that the desirable moisture level is reached, after that it will stop.

Machine Learning Algorithms

Three classification algorithms have been adopted from the RRL experiment: Decision Tree, Random Forest, and K-Nearest Neighbor. The reason behind adopting these algorithms is that these algorithms have proven to be effective for the problem of classification in the IoT environment sensors [4][9][13], but not because there are no other available algorithms; in fact, because these algorithms are good enough to classify low-dimensional sensor-based data generated by Bio-Sync. The best among these three algorithms has been selected on the basis of their performance in accuracy, precision, and recall after training them simultaneously.

Hardware Components

All the hardware elements mentioned below make up the Bio-Sync physical tier. The selection of each of the hardware items was based on its technical attributes, compatibility with the NodeMCU ESP8266 platform, and its usefulness in environments where mushrooms are cultivated and where the humidity is comparatively high.

a. NodeMCU ESP8266 (Microcontroller)

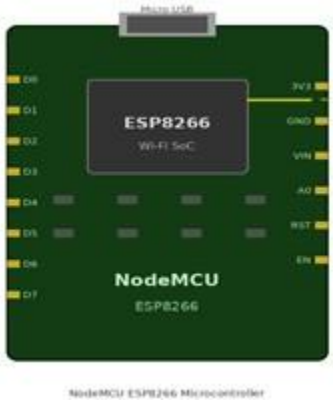


Figure 13. NodeMCU ESP8266 Microcontroller

Discussion: The NodeMCU ESP8266 acts as the processor and communications backbone of the whole hardware layer. This is a low-cost, Wi-Fi equipped microcontroller on the ESP8266 System-on-Chip, operating at 3.3V logic level and controlled through Arduino IDE utilizing the ESP8266 board support package. Since Wi-Fi is integrated into the SoC, an additional wireless adapter is unnecessary – this allows to simplify the wiring and reduce the number of parts required by the prototype. During its operation, NodeMCU periodically scans DHT22 sensor data at specified intervals, controls the execution of the threshold control algorithm for the current growth stage, controls the status of both relay ports and transfers telemetry to the web server in form of HTTP POST requests. GPIO pins (D0-D8) of NodeMCU are sufficient in quantity to cover all I/O connections that Bio-Sync requires, including the DHT22 output line and two relay control lines.

b. DHT22 Temperature and Humidity Sensor



Figure 14. DHT22 (AM2302) Temperature and Humidity Sensor

Discussion: DHT22 or commonly known as AM2302 is a capacitance-based relative humidity and NTC thermistor type temperature sensor which produces digital outputs via a serial interface through one wire connection. This device measures temperatures from −40°C to +80°C with an accuracy of ±0.5°C, while it also measures relative humidity levels from 0% to 100% RH with an accuracy level of ±2% to ±5% RH with a sampling rate of one measurement per two seconds. One unit of DHT22 sensor is installed inside the fruiting chamber in the middle of its height in order to measure ambient microclimate conditions. Performance specifications of this equipment completely meet the required monitoring parameters according to the literature-verified thresholds of Pleurotus ostreatus growth at all three growth stages [1][2][9]. In comparison to cheaper DHT11 sensor, DHT22 offers significantly higher precision and operating range which makes it the right choice for temperature and humidity control at narrow ranges which are vital for mushrooms cultivation process. The appropriateness of DHT22 has been confirmed by Othman et al. [1] and Kailashkumar et al. [2] studies.

c. 2-Channel 5V Optoisolated Relay Module

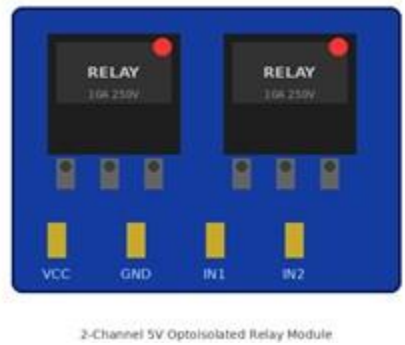


Figure 15. 2-Channel 5V Optoisolated Relay Module

Discussion: The 2-Channel Optoisolated Relay Module serves as the bridge from the NodeMCU's 3.3V control signal to the 12V DC loads that drive the actual fan and pump. The module allows each channel to be controlled by its own NodeMCU digital output pin, allowing the fan and pump to be individually or simultaneously operated based on the circumstances needed. The optoisolation is an important factor since it electrically isolates the control signal from the power that drives the actuator, providing protection from any possible voltage transient, inductive kickback, and electrical noise produced when the DC motors start and stop. The current ratings for each channel are 10A @ 250V AC and 10A @ 30V DC, which greatly exceeds what the system needs, and there are indicator LEDs in the module to help the team determine their status.

d. 12V DC Brushless Cooling Fan



Figure 16. 12V DC Brushless Cooling Fan

Discussion: The 12V DC brushless cooling fan is responsible for handling active ventilation. The relay turns on this fan if the temperature exceeds the upper limit of the specific phase of development, or when there is a need to provide fresh air circulation in order to avoid accumulation of CO2 and the absence of which results in elongation of stems and abnormal formation of caps. Brushless fans are preferred over brushed motors because of their increased lifespan, quietness and low maintenance needs in a humid environment. The 12V DC cooling fan is attached to the 12V power supply by means of Relay Channel 1 and all activities are logged in Actuator_Logs table.

e. 12V Mini Submersible Water Pump



Figure 17. 12V Mini Submersible Water Pump

Discussion: In contrast, the 12V mini submersible water pump deals with the reverse issue. The pump takes water from a separate reservoir and sprays it using the misting nozzle in order to increase relative humidity and re-hydrate the top layer of the growing medium. The pump is activated via the relay once the humidity level falls below the lower limit set for the respective growth phase. The submersibility of the pump is crucial because it allows to keep the device primed and ready to operate instantly without air locking, which is vital for the fast reaction that the closed-loop system requires. The pump operates using Relay Channel 2 and in order not to over-saturate the substrate, the closed loop logic of the system turns the pump off once the desired humidity level is achieved.

f. 12V DC Regulated Power Supply



Figure 18. 12V DC Regulated Power Supply

Discussion: Powering both actuators is a regulated 12V DC power supply, which can take a large range of AC input voltage (100V – 240V), making it suitable for use with regular Philippine household power (220V AC, 60Hz). It is essential that the power supply is regulated because variations in voltage may lead to actuator instability or even incorrect sensor readings. Both the NodeMCU and DHT22 are powered separately through a 5V micro-USB power adapter, connected to the USB port of NodeMCU, so that the sensor/microcontroller circuit is separated from the actuator circuit.

Software Stack

The software of Bio-Sync system consists of five layers, each layer dealing with a separate stage of data processing from hardware signal to operator screen.

a. Arduino IDE



Figure 19. Arduino IDE

Discussion: Firmware development happens in the Arduino IDE, chosen for its extensive library ecosystem, cross-platform support, and mature ESP8266 integration through the ESP8266 Arduino Core. Within the firmware, the IDE environment manages DHT22 polling through the Adafruit DHT library, controls the relay outputs via standard GPIO commands, implements the closed-loop threshold comparison logic, and structures the outgoing HTTP POST packets using the ESP8266WiFi and ESP8266HTTPClient libraries. Its serial monitor also gets heavy use during calibration, letting the team verify real sensor output and relay behavior before anything goes into the field.

b. Visual Studio Code



Figure 20. Visual Studio Code (VS Code)

Discussion: VS Code serves as the environment for development covering both firmware and web application codebase. Providing a workplace where the team can move between C++ firmware files, PHP backend scripts, and HTML/CSS/JS frontend files without switching editors.

c. HTML5, CSS3, and JavaScript (Frontend Web Interface)



Figure 21. HTML5, CSS3, and JavaScript

Discussion: The frontend runs on playing HTML5, CSS3, and JavaScript, purposely built without a heavier framework like React or Vue.js, to keep hosting simple and compatible with the low-cost shared hosting environments, which is typically available to community-scale farm operations. HTML5 provides the structural markup, CSS3 handles the styling and responsiveness of the dashboard across screen sizes, and JavaScript manages the dynamic behavior and communication through periodic AJAX polling of the PHP API.

d. PHP (Backend Application Layer)



Figure 22. PHP (Hypertext Preprocessor)

Discussion: On the server side, PHP scripts will play the role of an intermediary between NodeMCU, the MySQL database, and the frontend dashboard. The API endpoints get HTTP POST requests from the NodeMCU after every reading cycle, check the validity of the JSON payload being sent, and use the data to compare the readings with threshold values stored in the System_Settings table of the MySQL database, and update the System_Control table with the necessary state of actuators, before finally storing the timestamp in Sensor_Logs. Separate scripts will be responsible for serving the dashboard by querying the database and sending the data back in JSON format to the AJAX requests made by the front end. PHP was selected mostly because

of its maturity and compatibility with MySQL, as well as ease of deployment on shared hosting services.

e. MySQL (Database Management Engine)



Figure 23. MySQL Database

Discussion: MySQL is used as the underlying component layer for all of them; it is where all the data related to sensor telemetry, actuator logs, user accounts, thresholds, and notifications is stored within the six tables in a normalized form. The foreign key constraints are in place to ensure the referential integrity between the sensor reading and its associated device and also between the notification and the user and the device that triggered it. MySQL was selected for its reliability, availability, and ease of use on shared hosting services, its ACID properties, and its compatibility with PHP.

System Testing and Implementation

Before deployment into a live production mushroom chamber in Panabo City, the system must be run through a three-stage test sequence consisting of unit/component isolation test, integration test, and field acceptance test.

[UNIT / COMPONENT ISOLATION TEST] ⟶ [INTEGRATION TEST] ⟶ [FIELD ACCEPTANCE TEST]

1. Unit / Component Isolation Test: Each sensor node (DHT22 and moisture probes) is calibrated for base level accuracy using physical measurement instruments. Relays will be tested using continuous switch loops for thermal safety profile.
2. Integration Test: Communication is the focus of this phase as the reliable communication between microcontroller and the database APIs will be checked, as well as the speed of data processing, and actual actions performed by relays when the simulated sensor values exceed the threshold set by the program.
3. Field Acceptance Test: The entire hardware assembly will be installed inside the live mushroom fruiting chamber in Panabo City. Testing will include assessing the system's ability to cope with local cellular/Wi-Fi connection drops, physical durability in high humidity environment, and rapid response of the water pumps to dry substrate.

System Maintenance

The following maintenance regime is recommended for prolonging the lifespan of the automated system in the field:

- Hardware Maintenance Schedule: Deposits of minerals through spraying water may eventually affect the accuracy of sensors. The soil moisture sensors should be cleaned biweekly along with a monthly maintenance schedule to determine whether there is any corrosion or data corruption.

- Security and Software Patching Schedule: Updates will be conducted periodically to improve PHP file handling capabilities, to address vulnerabilities in the web dashboard access interface, and to make database backups using MySQL.
- Fail-Safe Protocol: In case of a failed sensor that went out of service or was disconnected, the default fail-safe routine is activated. The watering pump is disabled in order to prevent flooding, the hardware error is logged, and an alert message is sent to the owner's dashboard.

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